

GROUP 15

INTAKE AND EXHAUST

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GENERAL INFORMATION

M1151000101041

The exhaust pipe is divided into three parts.

SERVICE SPECIFICATION

M1151000301335

ITEM	STANDARD VALUE	LIMIT
Manifold distortion of the installation surface mm (in)	0.15 (0.006) or less	0.20 (0.008)
Intake charge pressure kPa (psi)	85 –159 (12.4 –23.0)	–
Turbocharger waste gate actuator pressure kPa (psi)	98 –102 (14.3 –14.7)	–
No. 1 Turbocharger waste gate solenoid terminal resistance [at 20° C (68° F)] Ω	29 –35	–
No. 2 Turbocharger waste gate solenoid terminal resistance [at 20° C (68° F)] Ω	29 –35	–
Exhaust manifold distortion of the installation surface mm (in)	–	0.70 (0.028)

INTAKE AND EXHAUST DIAGNOSIS**INTRODUCTION**

M1151006900406

Intake leaks usually create driveability issues that are not obviously related to the intake system. Exhaust leaks or abnormal noise is caused by cracks, gaskets and fittings, or by exhaust pipe or muffler damage due to impacts during travel. The exhaust leaks from these sections and causes the exhaust noise to increase. There may be cases when the system contacts the body and vibration noise is generated.

TROUBLESHOOTING STRATEGY

M1151007000398

Use these steps to plan your diagnostic strategy. If you follow them carefully, you will be sure that you have exhausted most of the possible ways to find an intake or exhaust system fault.

1. Gather information from the customer.

2. Verify that the condition described by the customer exists.
3. Find the malfunction by following the Symptom Chart.
4. Verify malfunction is eliminated.

SYMPTOM CHART

M1151007100395

Symptom	Inspection procedure	Reference page
Exhaust Leakage	1	P.15-3
Abnormal Noise	2	P.15-3

SYMPTOM PROCEDURES

Inspection Procedure 1: Exhaust Leakage

DIAGNOSIS

STEP 1. Start the engine. Have an assistant stay in the driver's seat. Raise the vehicle on a hoist. Have the assistant rev the engine while searching for exhaust leaks.

Q: Is the exhaust leaking?

YES : Go to Step 2.

NO : The procedure is complete.

STEP 2. Check the gasket for cracks, damage.

Q: Is the gasket damaged?

YES : Replace the gasket, then go to Step 1.

NO : Go to Step 3.

STEP 3. Check for loosening in each coupling section.

Q: Is there any loosening in any section?

YES : Tighten, then go to Step 1.

NO : There is no action to be taken.

Inspection Procedure 2: Abnormal Noise

DIAGNOSIS

STEP 1. Start the engine. Have an assistant stay in the drivers seat. Raise the vehicle on a hoist. Have the assistant rev the engine while searching for exhaust leaks.

Q: Is any abnormal noise generated?

YES : Go to Step 2.

NO : The procedure is complete.

STEP 2. Check for missing parts in the muffler. Tap the muffler lightly to check for loose baffles, etc.

Q: Are there any missing parts in the muffler?

YES : Replace, then go to Step 1.

NO : Go to Step 3.

STEP 3. Check the hanger for cracks.

Q: Is the hanger cracked?

YES : Replace, then go to Step 1.

NO : Go to Step 4.

STEP 4. Check for interference of the pipes and muffler with the body.

Q: Are the pipes and muffler interfering with the body?

YES : Repair, then go to Step 1.

NO : Go to Step 5.

STEP 5. Check the heat protectors.

Q: Are any heat protectors loose or damaged?

YES : Tighten or replace, then go to Step 1.

NO : Go to Step 6.

STEP 6. Check the pipes and muffler for damage.

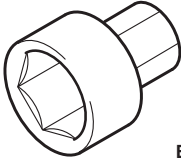
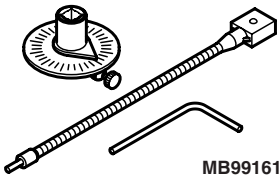
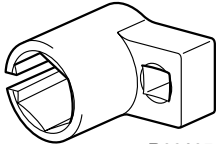
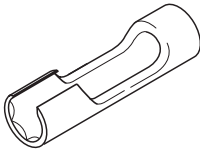
Q: Are the pipes and muffler damaged?

YES : Replace, then go to Step 1.

NO : There is no action to be taken.

SPECIAL TOOLS

M1151000601024

Tool	Tool number and name	Supersession	Application
 B992274	MB992274 Palm socket	-	Removal and installation of turbocharger compressor bracket
 MB991614	MB991614 Angle gauge	General service tool	Installation of turbocharger assembly coupling bolt and nut
 B991953	MB991953 Oxygen sensor wrench	MB991953-01	Removal and installation of heated oxygen sensor
	MB992188 Fuel injection pipe wrench	MB992188-01	Removal and installation of No.2 air temperature sensor

TROUBLESHOOTING

M1151010200035

Symptom	Probable cause	Remedy
Exhaust gas leakage	Loose joints	Retighten
	Broken pipe or muffler	Repair or replace
Abnormal noise	Broken baffle in muffler	Replace
	Broken rubber hangers	Replace
	Interference of pipe or muffler with vehicle body	Correct
	Broken pipe or muffler	Repair or replace

ON-VEHICLE SERVICE

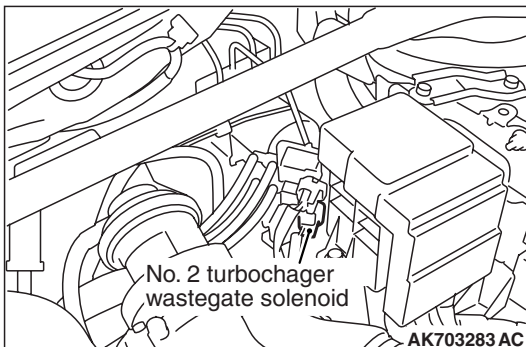
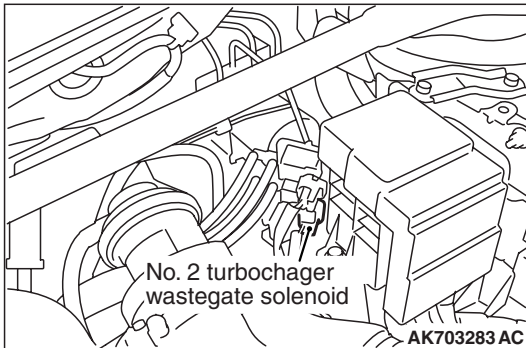
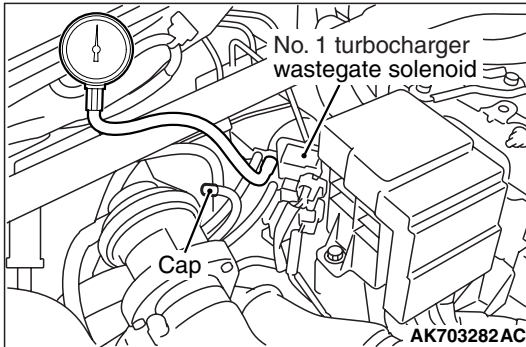
MANIFOLD VACUUM CHECK

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Refer to GROUP 11A, Engine Mechanical –On-vehicle Service
–Manifold Vacuum Check. [P.11A-16](#)

INTAKE CHARGE PRESSURE CHECK

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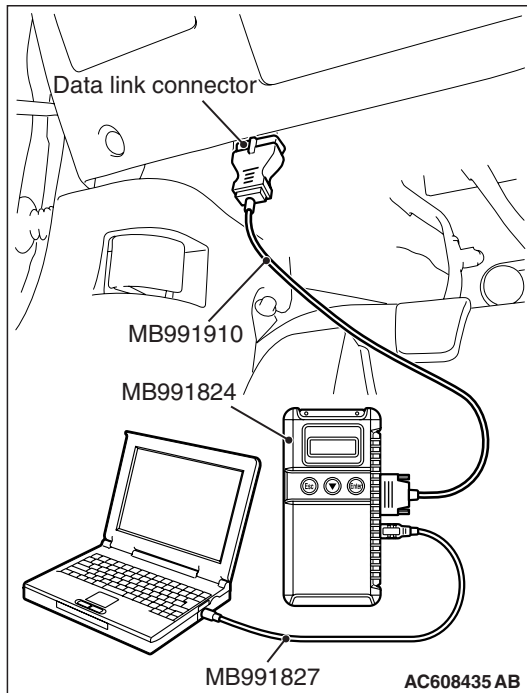


1. Removing the black hose from the No.1 turbocharger wastegate solenoid or the brown connector, install the pressure gauge to this hose. Cap the nipple of solenoid after removing the black hose.

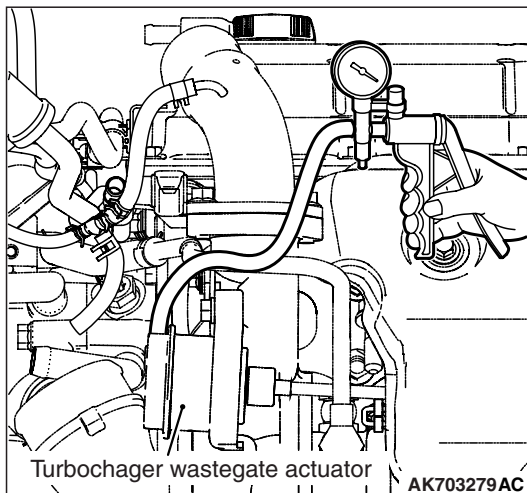
2. Disconnect the black connector of No.2 turbocharger wastegate solenoid.
3. When the engine speed reaches approximately 3000r/min or more with the throttle fully opened in the second gear, measure the boost pressure.

Standard value: 85 –159kPa (12.4 –23.0 psi)

4. If the boost pressure is lower than the standard, check the following possible causes.
 - a. Turbocharger wastegate actuator malfunction
 - b. Boost pressure leakage
 - c. Turbocharger malfunction
5. If the boost pressure is higher than the standard, check on the following areas because the boost pressure control might be abnormal.
 - a. Turbocharger wastegate actuator malfunction
 - b. Wastegate regulating valve malfunction
 - c. Rubber hose of turbocharger wastegate actuator disconnected or cracked
6. Connect the black connector of No.2 turbocharger wastegate solenoid.



7. If the diagnostic trouble code is stored during the check, use scan tool MB991958 to eliminate the diagnosis code.



TURBOCHARGER WASTEGATE ACTUATOR CHECK

M1151001200413

1. Connect a hand vacuum pump (pressure-application type) to nipple.

⚠ CAUTION

In order to avoid damage to the diaphragm, do not apply a pressure of 117 kPa (17 psi) or higher.

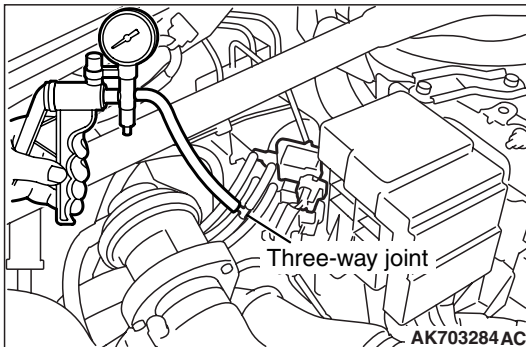
2. While gradually applying pressure, check the pressure that begins to activate (approximately 1 mm stroke) the wastegate actuator rod.

Standard value: 98 –102 kPa (14.3 –14.7 psi)

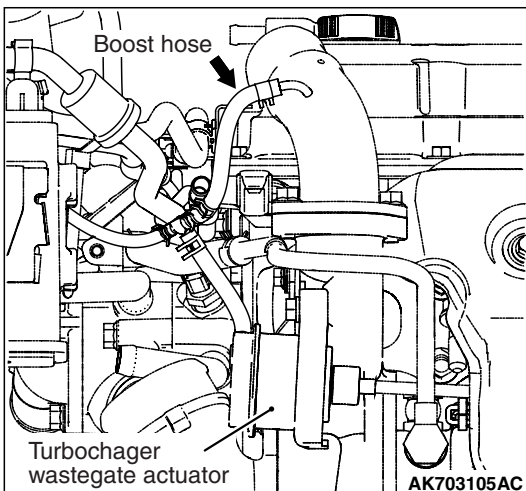
3. If there is a significant deviation from the standard value, check the actuator or the wastegate valve: replace if necessary.

INTAKE CHARGE PRESSURE CONTROL SYSTEM CHECK

M1151001100342



1. Removing the black hose from the No.1 turbocharger wastegate solenoid or the brown connector, connect three-way joint between the hose and the solenoid.
2. Connect the pressurized type hand vacuum pump with the three-way joint.



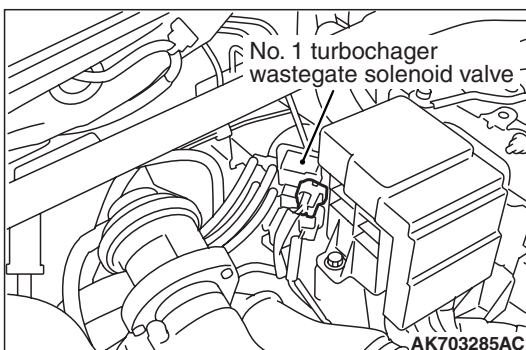
3. Use the turbocharger wastegate actuator control boost nipple of air outlet fitting to remove the boost hose. Plug this nipple.

CAUTION

Do not apply the pressure of 117kPa or more to prevent the diaphragm damage.

4. Blocking or releasing the boost hose end, apply the pressure. Check the pressurized state.

Engine state	Boost hose end	Normal state
Stopped: Ignition switch in "ON" position	Opened	Pressure leaks
	Closed	Pressure maintained
Rapid racing		Pressure leaks



5. Put the ignition switch in "LOCK" (OFF) position.
6. Disconnect the connector of No.1 turbocharger wastegate solenoid or the brown connector.
7. Blocking or releasing the boost hose end, apply the pressure. Check the pressurized state. Plug the boost hose end while driving.

Engine state	Boost hose end	Normal state
Stopped: Ignition switch in "ON" position	Opened	Pressure leaks
	Closed	Pressure maintained
4000r/min after warming up engine		Pressure leaks

NOTE: Unless the pressurized state is normal, the turbocharger wastegate actuator, turbocharger wastegate solenoid or the hose might have a malfunction.

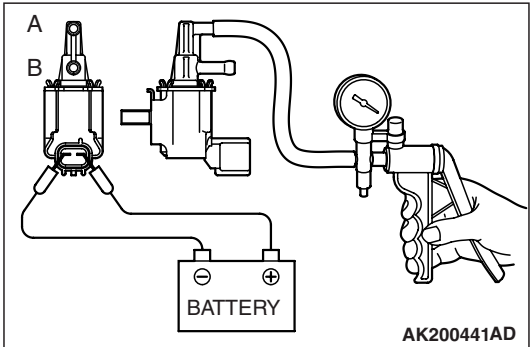
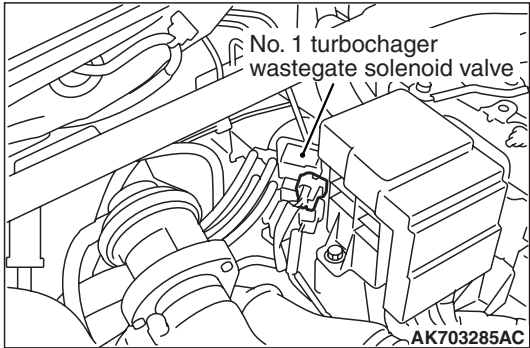
TURBOCHARGER WASTE GATE SOLENOID
CHECK

M1151001300272

No.1 turbocharger wastegate solenoid or brown
connector

A Operation check

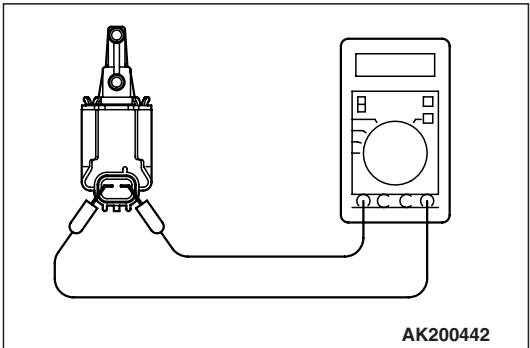
1. Install the hand vacuum pump to the nipple "A" of solenoid.
2. Use the jumper wire to connect the solenoid terminal with the battery terminal.
3. Disconnecting the jumper wire at the negative (-) battery side, apply the vacuum pressure. Check the airtightness.



Jumper wire	Nipple "B" state	Normal state
Connected	Opened	Negative pressure leaks
	Closed	Negative pressure maintained
Disconnected	Opened	Negative pressure maintained

B Check of coil resistance

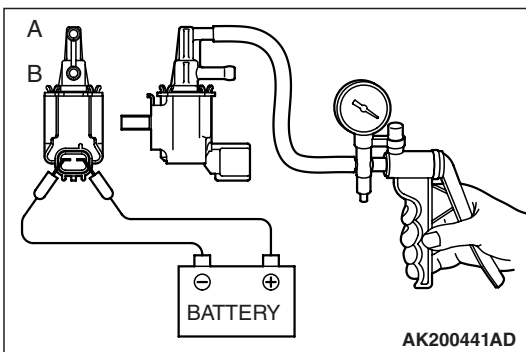
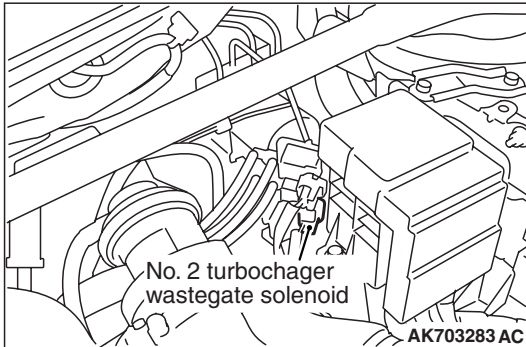
1. Measure the resistance between the solenoid terminals.
Standard value: 29 –35 Ω at 20° C



NO.2 turbocharger wastegate solenoid or black connector

A Operation check

1. Install the hand vacuum pump to the nipple "A" of solenoid.
2. Use the jumper wire to connect the solenoid terminal with the battery terminal.
3. Disconnecting the jumper wire at the negative (-) battery side, apply the vacuum pressure. Check the airtightness.

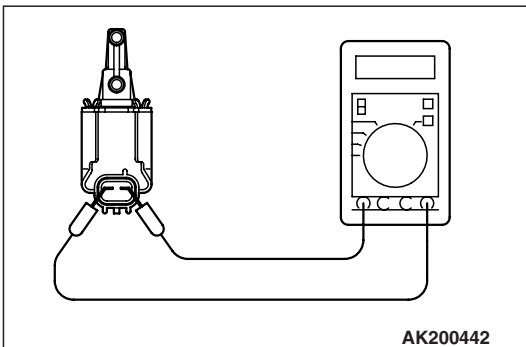


Jumper wire	Nipple "B" state	Normal state
Connected	Opened	Negative pressure leaks
	Closed	Negative pressure maintained
Disconnected	Opened	Negative pressure maintained

B Check of coil resistance

1. Measure the resistance between the solenoid terminals.

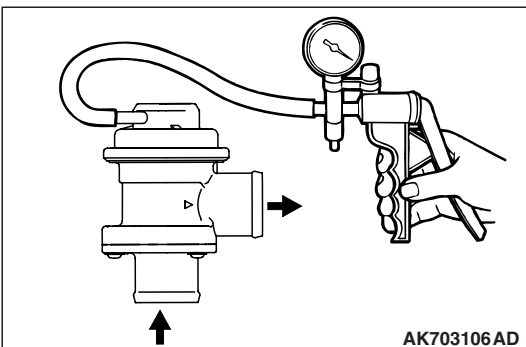
Standard value: 29 –35 Ω at 20° C



TURBOCHARGER BYPASS VALVE CHECK

M1151001600217

1. Remove the turbocharger bypass valve.
2. Connect the hand vacuum pump to the nipple of the turbocharger bypass valve.
3. Apply a negative pressure of approximately 65 kPa (19 in.Hg) and check operation of the valve. Also check that air tightness is maintained.

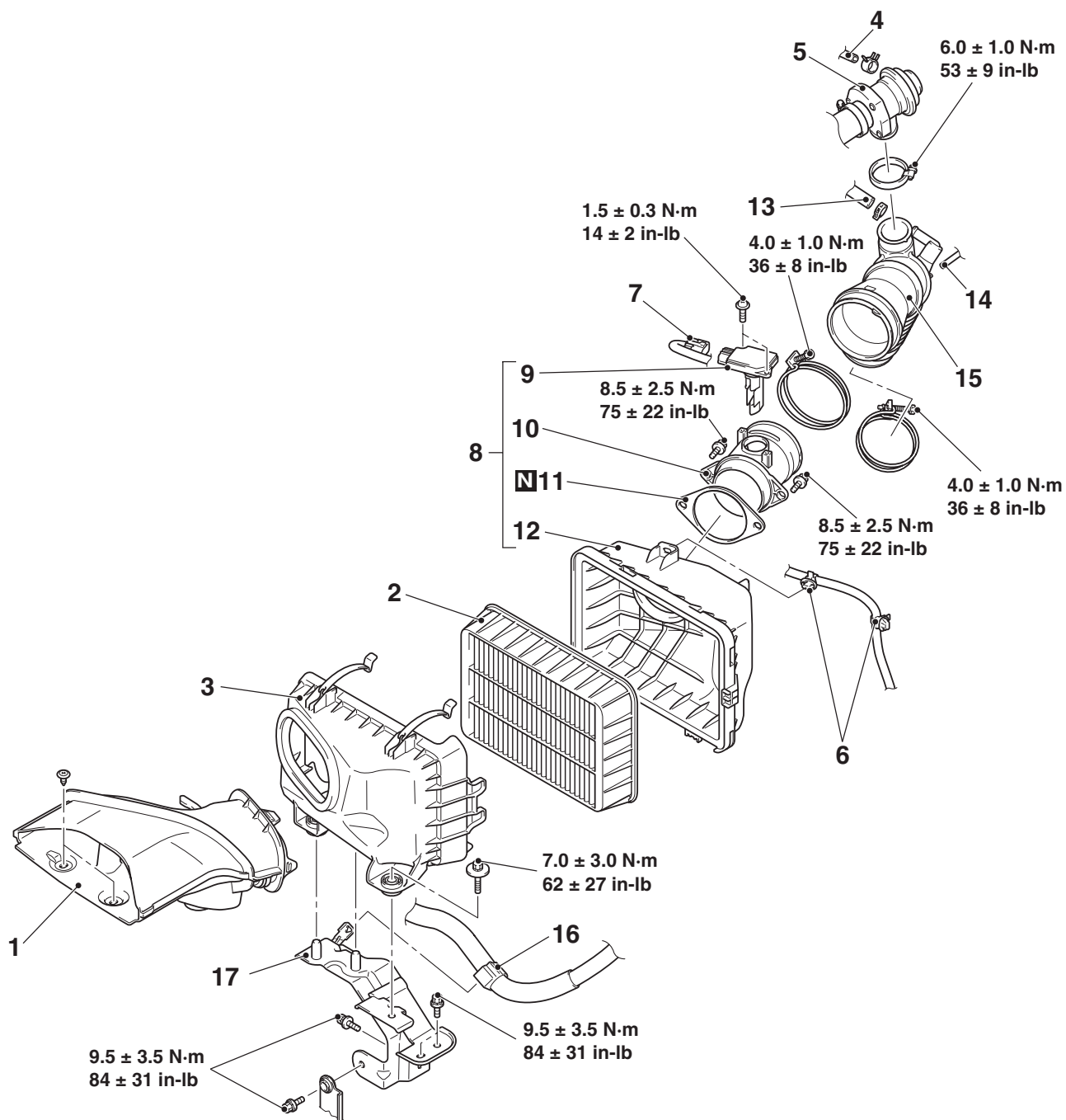


Negative pressure	Valve operation
Approximately 65 kPa (19 in.Hg)	Starts opening

AIR CLEANER

REMOVAL AND INSTALLATION

M1151002101951



AC710665 AB

Removal steps

1. Air cleaner intake duct
2. Air cleaner element
3. Air cleaner body
4. Air by-pass valve vacuum hose connection
5. Air by-pass valve connection
6. Harness clamp connection
7. Mass airflow sensor connector connection
8. Air cleaner cover assembly

>>A<<

Removal steps (Continued)

9. Mass airflow sensor
10. Air cleaner support
11. Gasket
12. Air cleaner cover
13. Breather hose connection
14. Vacuum hose connection
15. Air cleaner intake hose
16. Harness clamp connection
17. Air cleaner bracket

INSTALLATION SERVICE POINT

>>A<< MASS AIRFLOW SENSOR INSTALLATION

 CAUTION

Do not over-tighten. If the mass airflow sensor installation screw is over-tightened, the air cleaner support thread may be damaged.

INTAKE MANIFOLD

REMOVAL AND INSTALLATION

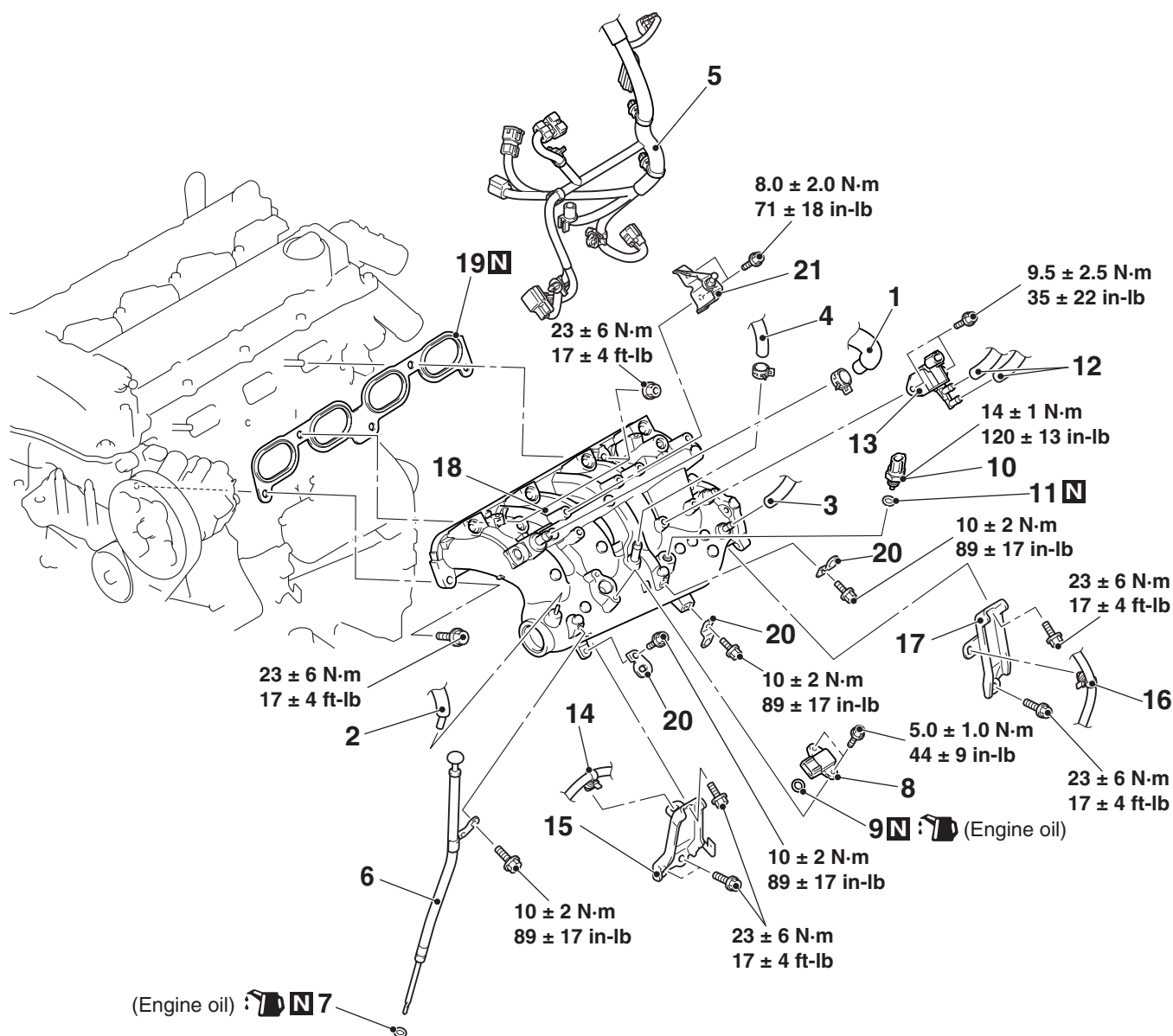
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Pre-removal operation

- Engine Upper Cover Removal (Refer to GROUP 16, Ignition System –Ignition Coil [P.16-39](#)).
- Air By-pass Valve, Air By-pass Hose, Charge Air Cooler Intake Hose B Removal (Refer to [P.15-15](#)).
- Air Cleaner Assembly Removal (Refer to [P.15-10](#)).
- Drive Belt Removal (Refer to GROUP 11A, Crankshaft Pulley [P.11A-21](#)).
- Throttle Body Removal (Refer to GROUP 13A, Throttle Body [P.13A-918](#)).
- Fuel Delivery Pipe and Fuel Injector Assembly Removal (Refer to GROUP 13A, Injector [P.13A-914](#)).

Post-installation operation

- Fuel Delivery Pipe and Fuel Injector Assembly Installation (Refer to GROUP 13A, Injector [P.13A-914](#)).
- Throttle Body Installation (Refer to GROUP 13A, Throttle Body [P.13A-918](#)).
- Air Cleaner Assembly Installation (Refer to [P.15-10](#)).
- Drive Belt Installation (Refer to GROUP 11A, Crankshaft Pulley [P.11A-21](#)).
- Air By-pass Valve, Air By-pass Hose, Charge Air Cooler Intake Hose B Installation (Refer to [P.15-15](#)).
- Engine Upper Cover Installation (Refer to GROUP 16, Ignition System –Ignition Coil [P.16-39](#)).



AC711118AC

Removal steps

1. Rocker cover PCV hose
2. Emission control equipment vacuum hose connection

Removal steps (Continued)

3. Emission vacuum hose connection
4. Brake booster vacuum hose connection

Removal steps (Continued)

5. Control wiring harness connection
6. Engine oil level gauge
7. O-ring
- >>B<< 8. Manifold absolute pressure sensor
- >>B<< 9. O-ring
- <<A>> >>A<< 10. No.2 Intake air temperature sensor
11. Gasket
12. Vacuum hose connection
13. Purge control solenoid valve
14. Starter wiring harness clamp
15. Intake manifold stay (front)
16. Starter wiring harness clamp
17. Intake manifold stay (rear)
18. Intake manifold
19. Intake manifold gasket
20. Harness bracket
21. Engine upper cover bracket (front)

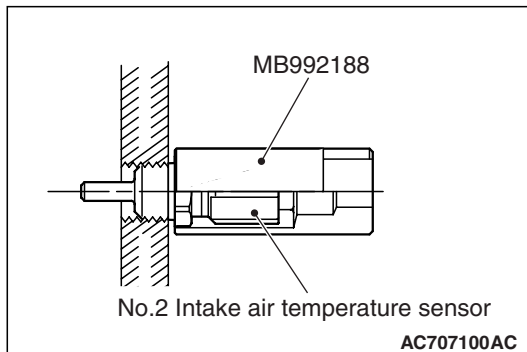
Required Special Tool:

- MB992188: Fuel Injection Pipe Wrench

REMOVAL SERVICE POINT

<<A>> NO.2 INTAKE AIR TEMPERATURE SENSOR REMOVAL

Use special tool MB992188 to remove the No.2 intake air temperature sensor.

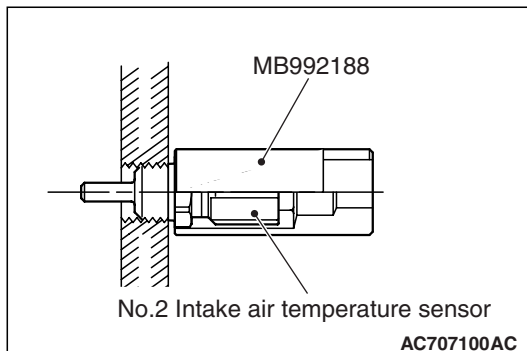


INSTALLATION SERVICE POINTS

>>A<< NO.2 INTAKE AIR TEMPERATURE SENSOR INSTALLATION

Tighten the No.2 intake air temperature sensor to the specified torque by using special tool MB992188.

Tightening torque: 14 ± 1 N·m (120 ± 13 in-lb)



**>>B<< O-RING/MANIFOLD ABSOLUTE
PRESSURE SENSOR INSTALLATION**** CAUTION**

- Install the manifold absolute pressure sensor, taking care that no shock is applied to it.
 - Do not use a manifold absolute pressure sensor that has been dropped.
1. Apply a small amount of new engine oil to the O-ring.
 2. Install the manifold absolute pressure sensor to the intake manifold with care not to damage the O-ring.

INSPECTION

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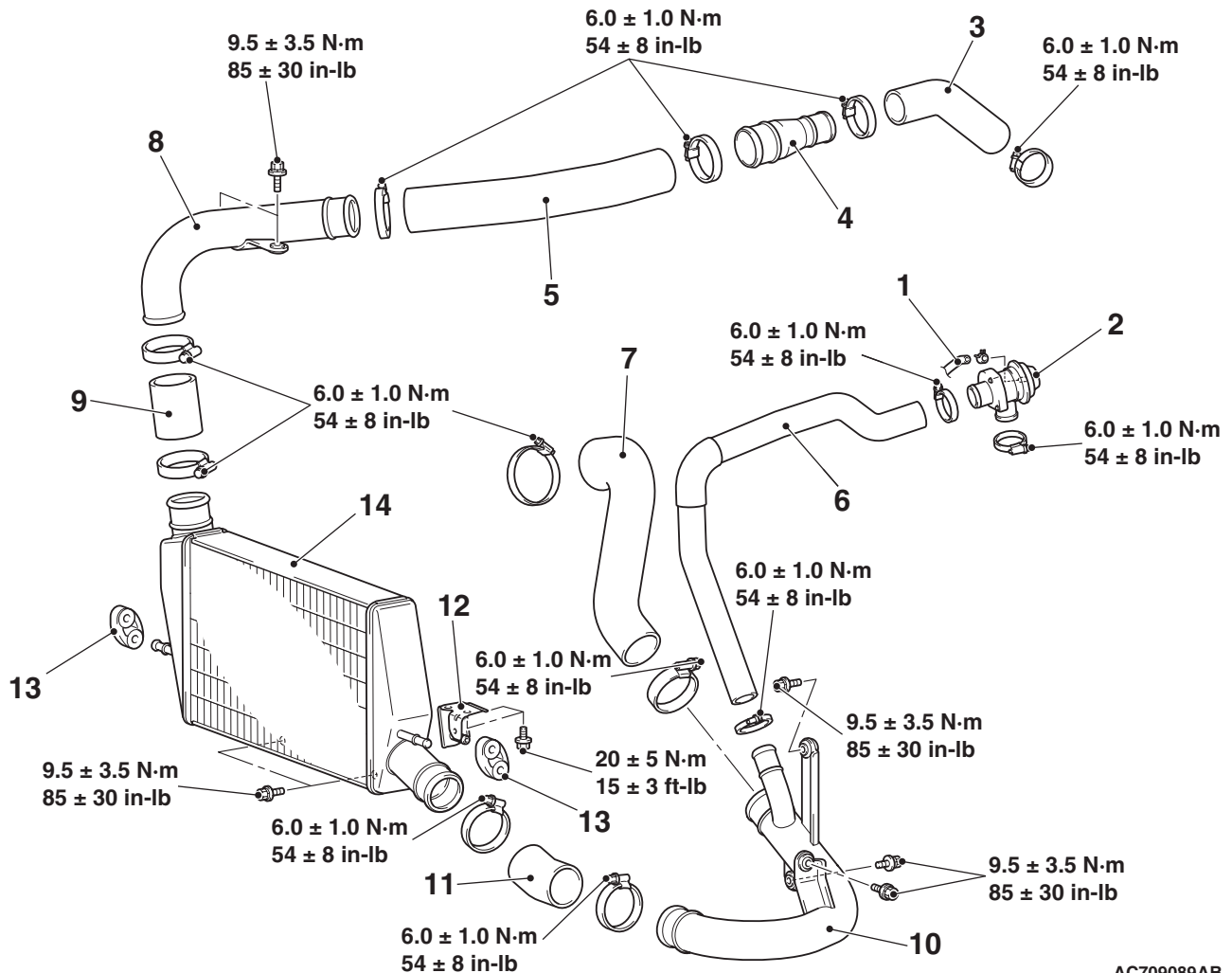
INTAKE MANIFOLD CHECK

1. Check the intake manifold for damage and cracks, and replace it if necessary.
2. Check the vacuum outlet port for clogging, and clean it if necessary.

CHARGE AIR COOLER

REMOVAL AND INSTALLATION

M1151002400410



AC709089AB

Removal steps

1. Turbocharger by-pass valve vacuum hose connection
2. Turbocharger by-pass valve
3. Charge air cooler intake hose A
4. Charge air cooler intake pipe A
5. Charge air cooler intake hose B
 - Air cleaner body (Refer to P.15-10)
6. Turbocharger by-pass valve hose
7. Charge air cooler outlet hose E
 - Headlight support panel cover (Refer to GROUP 51, Front Bumper and Radiator Grille Assembly P.51-16)
8. Charge air cooler intake pipe B

Removal steps (Continued)

9. Charge air cooler intake hose D
 - Front bumper and radiator grille assembly (Refer to GROUP 51, Front Bumper and Radiator Grille Assembly P.51-16)
10. Charge air cooler outlet pipe C
11. Charge air cooler outlet hose C
12. Charge air cooler hanger bracket (LH)
13. Charge air cooler hanger
14. Charge air cooler assembly

EXHAUST MANIFOLD AND TURBOCHARGER

REMOVAL AND INSTALLATION

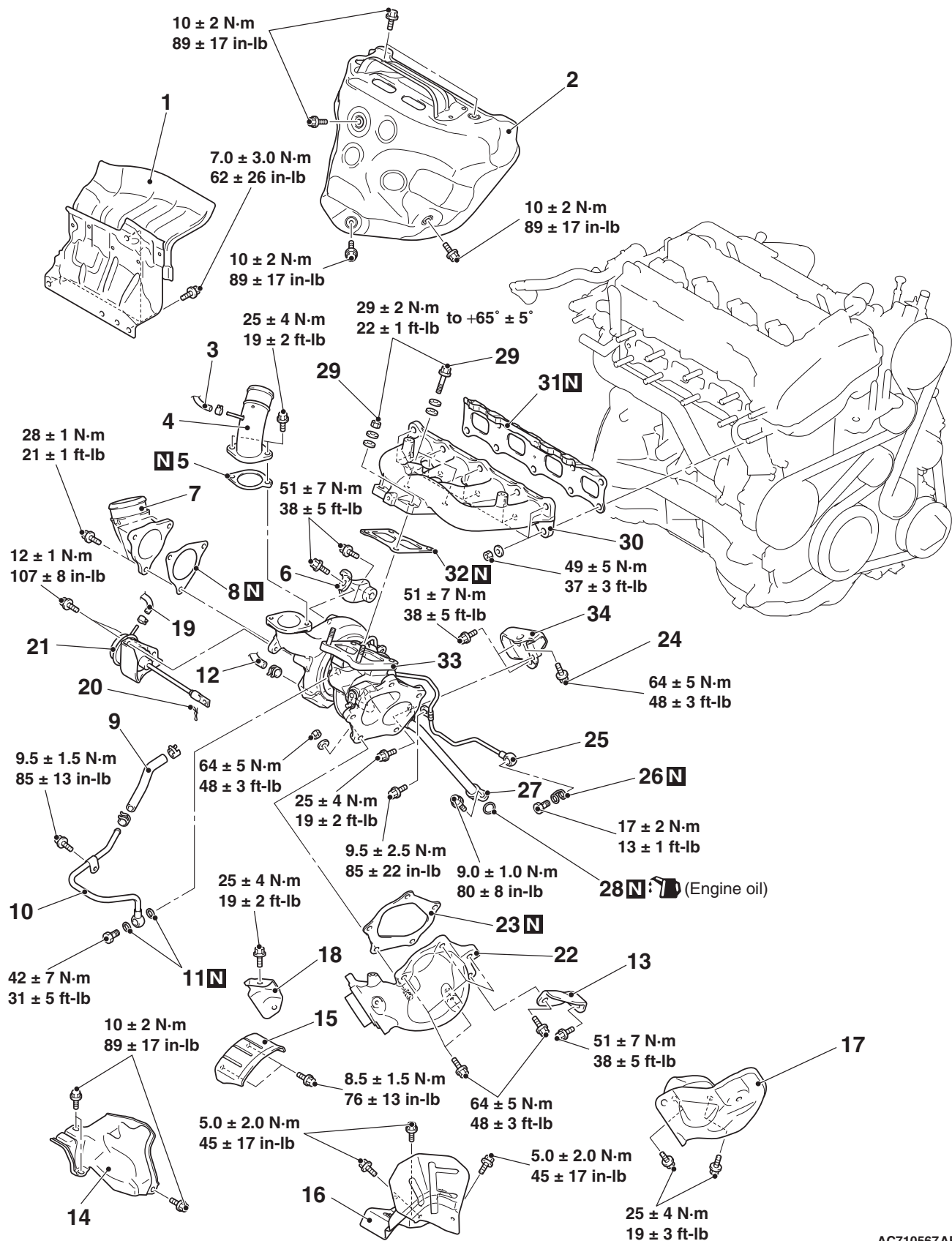
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Pre-removal operation

- Engine Room Under Cover Front A, B, Engine Room Under Cover Center and Engine Room Side Cover (RH) Removal (Refer to GROUP 51, Under Cover [P.51-16](#)).
- Engine Coolant Draining (Refer to GROUP 14, On-vehicle Service –Engine Coolant Replacement [P.14-17](#)).
- Engine Oil Draining (Refer to GROUP 12, On-vehicle Service –Engine Oil Replacement [P.12-5](#)).
- Engine Upper Cover Removal (Refer to GROUP 16, Ignition System –Ignition Coil [P.16-39](#)).
- Charge Air Cooler Intake Hose A and Charge Air Cooler Intake Pipe A Removal (Refer to [P.15-15](#)).
- Air Cleaner Assembly and Air Cleaner Intake Hose Removal (Refer to [P.15-10](#)).
- Front Exhaust Pipe Removal (Refer to [P.15-24](#)).
- Strut Tower Bar Removal (Refer to GROUP 42A, Strut Tower Bar [P.42A-15](#)).
- Cowl Top Panel Removal (Refer to GROUP 42A, Loose Panel [P.42A-15](#)).

Post-installation operation

- Cowl Top Panel Installation (Refer to GROUP 42A, Loose Panel [P.42A-15](#)).
- Strut Tower Bar Installation (Refer to GROUP 42A, Strut Tower Bar [P.42A-15](#)).
- Front Exhaust Pipe Installation (Refer to [P.15-24](#)).
- Charge Air Cooler Intake Hose A and Charge Air Cooler Intake Pipe A Installation (Refer to [P.15-15](#)).
- Air Cleaner Assembly and Air Cleaner Intake Hose Installation (Refer to [P.15-10](#)).
- Engine Oil Refilling (Refer to GROUP 12, On-vehicle Service –Engine Oil Replacement [P.12-5](#)).
- Engine Coolant Refilling (Refer to GROUP 14, On-vehicle Service –Engine Coolant Replacement [P.14-17](#)).
- Engine Upper Cover Removal (Refer to GROUP 16, Ignition System –Ignition Coil [P.16-39](#)).
- Engine Room Under Cover Front A, B, Engine Room Under Cover Center and Engine Room Side Cover (RH) Removal (Refer to GROUP 51, Under Cover [P.51-16](#)).



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Removal steps

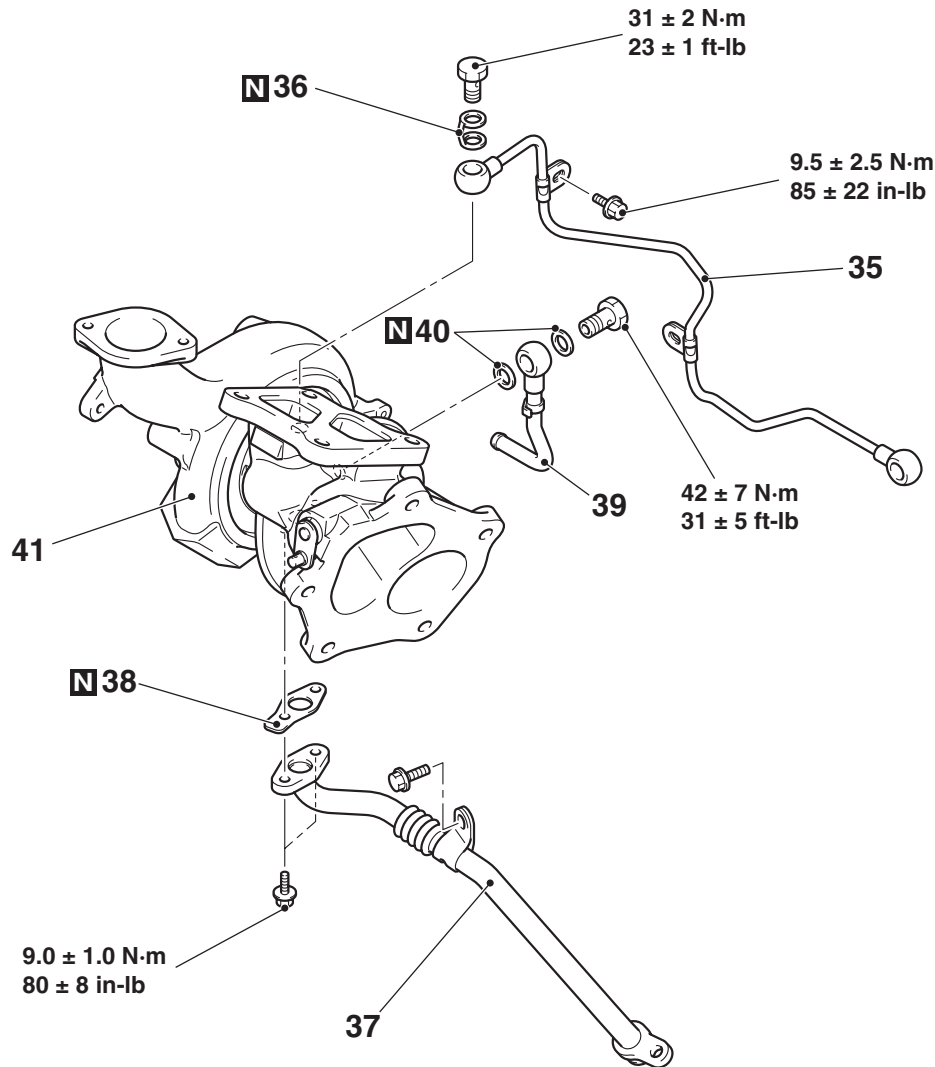
1. Dash panel heat protector
2. Exhaust manifold cover

Removal steps (Continued)

3. Emission vacuum control hose connection
4. Turbocharger air outlet fitting

Removal steps (Continued)

- | | | |
|-------|-------|---|
| | 5. | Turbocharger air outlet fitting gasket |
| <<A>> | >>F<< | 6. Turbocharger compressor bracket |
| | 7. | Turbocharger air inlet fitting |
| | 8. | Turbocharger air inlet fitting gasket |
| | 9. | Turbocharger water feed hose |
| | 10. | Turbocharger water feed pipe |
| | 11. | Gasket |
| | 12. | Turbocharger water return hose |
| >>E<< | 13. | Turbocharger exhaust outlet fitting bracket |
| | 14. | Transfer heat protector |
| | 15. | Drive shaft heat protector |
| | 16. | Steering gear and linkage heat protector |
| | 17. | Turbocharger protector A |
| | 18. | Turbocharger protector B |
| | 19. | Emission vacuum control hose connection |
| <> | 20. | Turbocharger pin |
| | 21. | Waste gate actuator |
| | 22. | Turbocharger exhaust outlet fitting |
| | 23. | Turbocharger exhaust outlet fitting gasket |
| >>C<< | 24. | Turbocharger bracket and turbocharger assembly coupling bolt |
| | 25. | Turbocharger oil feed tube connection |
| | 26. | Gasket |
| | 27. | Turbocharger oil return tube connection |
| | 28. | O-ring |
| >>D<< | 29. | Exhaust manifold and turbocharger assembly coupling bolt and nut |
| | 30. | Exhaust manifold |
| | 31. | Exhaust manifold gasket |
| | 32. | Turbocharger gasket |
| | 33. | Turbocharger assembly |
| | • | Transfer assembly (Refer to GROUP 22A, Transfer Assembly P.22A-125 <M/T>), (Refer to GROUP 22C, Transfer Assembly P.22C-339 <TC-SST>) |
| >>C<< | 34. | Turbocharger bracket |



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- Removal steps**
- <<C>>
- 35. Turbocharger oil feed tube
 - 36. Gasket
- >>B<<
- 37. Turbocharger oil return tube
 - 38. Turbocharger oil return tube gasket

- Removal steps (Continued)**
- >>A<<
- 39. Turbocharger water return pipe
 - 40. Gasket
 - 41. Turbocharger

Required Special Tools:

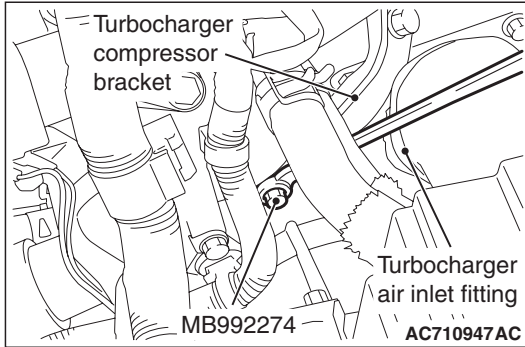
- MB992274: Palm Socket

- MB991614: Angle Gauge

REMOVAL SERVICE POINTS

<<A>> TURBOCHARGER COMPRESSOR BRACKET REMOVAL

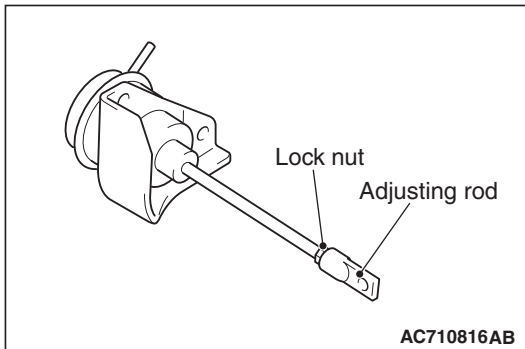
Use special MB992274 to remove the turbocharger compressor bracket mounting bolt (cylinder block side).



<> WASTE GATE ACTUATOR REMOVAL

CAUTION

Never loosen the locking nuts and adjusting rod of the waste gate actuator.



<<C>> TURBOCHARGER OIL FEED TUBE REMOVAL

CAUTION

Take care not to allow foreign objects to get into the oil passage hole of the turbocharger assembly after the turbocharger oil feed tube is removed.

INSTALLATION SERVICE POINTS

>>A<< TURBOCHARGER INSTALLATION

1. Clean the turbocharger oil feed tube, turbocharger water feed pipe and turbocharger water return pipe fitting, the inside of eye bolts, and individual pipe for clogs.

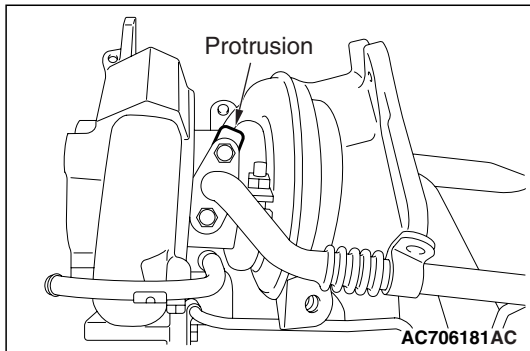
CAUTION

Take care not to allow foreign objects to get into the turbocharger.

2. Clean or use compressed air to remove any carbon particles stuck to the oil passage of the turbocharger.
3. Refill new engine oil at the turbocharger oil feed tube fitting hole of the turbocharger.

>>B<< TURBOCHARGER OIL RETURN TUBE GASKET INSTALLATION

Install the gasket as its protrusion is in the direction shown. Install the gasket so that its protrusion faces in the direction shown in the illustration.



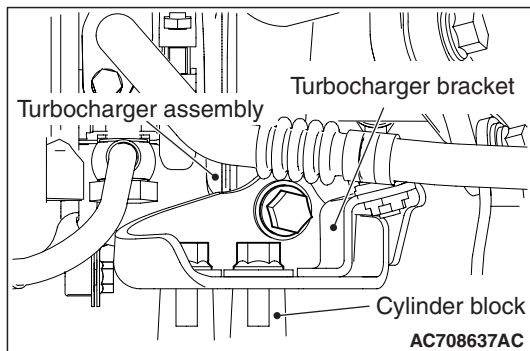
>>C<< TURBOCHARGER BRACKET/TURBOCHARGER BRACKET AND TURBOCHARGER ASSEMBLY COUPLING BOLT INSTALLATION

1. Check that the turbocharger bracket is in close contact with the turbocharger assembly and with the cylinder block, and then temporarily tighten the turbocharger bracket.
2. Tighten the bolt of the cylinder block side to the specified torque.

Tightening torque: 51 ± 7 N·m (38 ± 5 ft-lb)

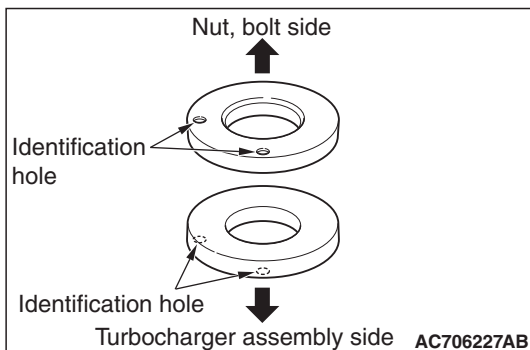
3. Tighten the turbocharger bracket and the turbocharger assembly coupling bolt to the specified torque.

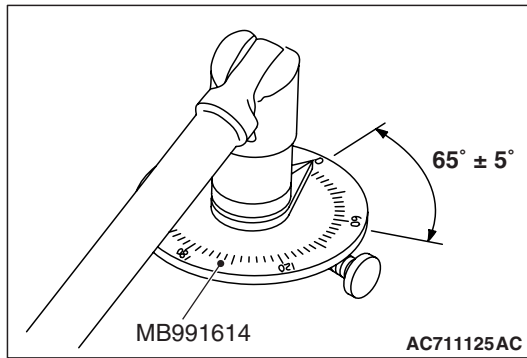
Tightening torque: 64 ± 5 N·m (48 ± 3 ft-lb)



>>D<< EXHAUST MANIFOLD AND TURBOCHARGER ASSEMBLY COUPLING BOLT AND NUT INSTALLATION

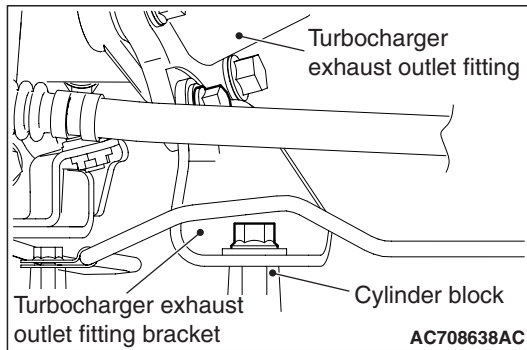
1. Tighten the bolts and nuts according to the procedure below.
 - (1) Install two washers with their identification holes facing outside as shown in the figure.
 - (2) Tighten the bolts and nuts to 29 ± 2 N·m (22 ± 1 ft-lb) in the order of number shown in the figure.





- (3) Use special tool MB991614 to tighten bolts and nuts $65^\circ \pm 5^\circ$.

>>E<< TURBOCHARGER EXHAUST OUTLET FITTING BRACKET INSTALLATION



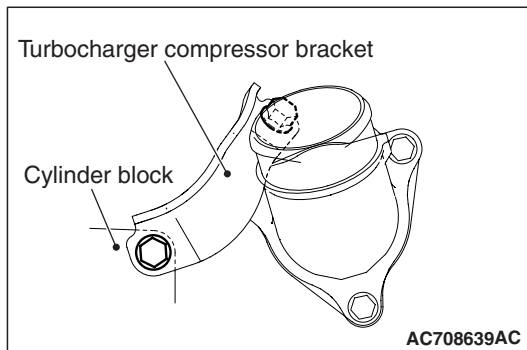
1. Check that the turbocharger exhaust outlet fitting bracket is in close contact with the turbocharger exhaust outlet fitting and with the cylinder block, and then temporarily tighten the turbocharger exhaust outlet fitting bracket.
2. Tighten the bolt of the cylinder block side to the specified torque.

Tightening torque: $51 \pm 7 \text{ N} \cdot \text{m}$ ($38 \pm 5 \text{ ft-lb}$)

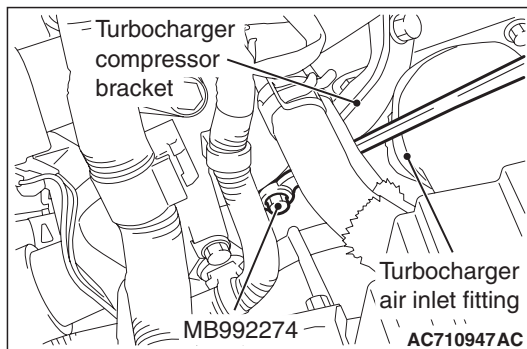
3. Tighten the bolt of the turbocharger exhaust outlet fitting bracket side to the specified torque.

Tightening torque: $64 \pm 5 \text{ N} \cdot \text{m}$ ($48 \pm 3 \text{ ft-lb}$)

>>F<< TURBOCHARGER COMPRESSOR BRACKET INSTALLATION



1. Check that the turbocharger compressor bracket is in close contact with the turbocharger assembly and with the cylinder block, and then temporarily tighten the turbocharger compressor bracket.



2. Use special tool MB992274 to tighten the turbocharger compressor bracket mounting bolt (cylinder block side) to the specified torque.

Tightening torque: $51 \pm 7 \text{ N} \cdot \text{m}$ ($38 \pm 5 \text{ ft-lb}$)

3. Tighten the bolt of the turbocharger assembly side to the specified torque.

Tightening torque: $51 \pm 7 \text{ N} \cdot \text{m}$ ($38 \pm 5 \text{ ft-lb}$)

INSPECTION

M1151009400132

EXHAUST MANIFOLD CHECK

1. Check the exhaust manifold for damage and cracks, and replace it if necessary.
2. Using a straight edge and a thickness gauge, check for distortion of the cylinder head and turbocharger installation surface. If it exceeds the limit value, replace it.

**Limit <Cylinder head assembly side>: 0.70 mm
(0.028 inch)**

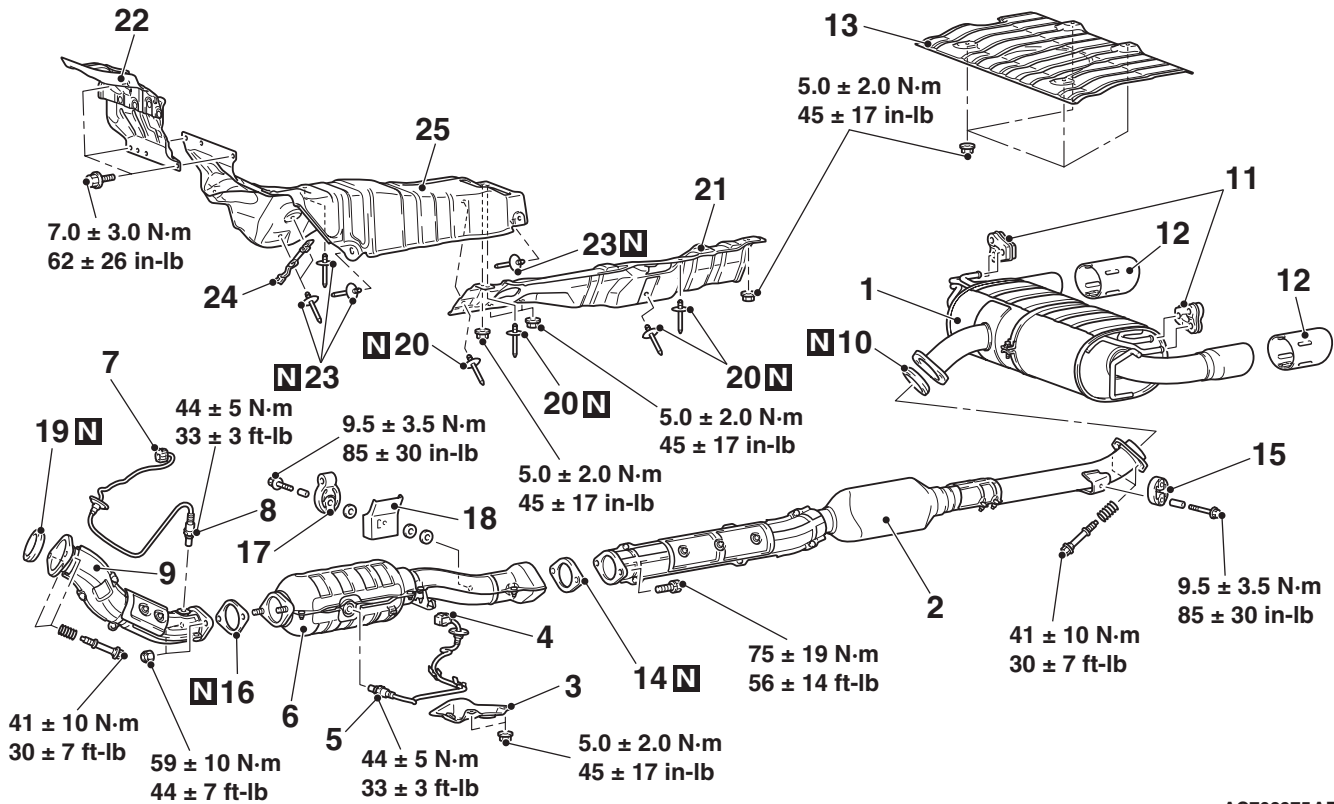
**Limit <Turbocharger assembly side>: 0.40 mm (0.016
inch)**

TURBOCHARGER ASSEMBLY CHECK

1. Visually check the turbine wheel and the compressor wheel for cracking or other damage.
2. Check whether the turbine wheel and the compressor wheel can be easily turned by hand.
3. Check for oil leakage from the turbocharger assembly.
4. Check whether or not the turbocharger waste gate regulating valve remains open. If any problem is found, replace the part after disassembly.

EXHAUST PIPE AND MAIN MUFFLER**REMOVAL AND INSTALLATION**

M1151008701854



AC708975AB

Exhaust main muffler and rear floor panel heat protector removal steps

1. Exhaust main muffler
10. Seal ring
11. Exhaust muffler hanger
12. Exhaust tail pipe diffuser
13. Rear floor panel heat protector

Center exhaust pipe removal steps

2. Center exhaust pipe
10. Seal ring
14. Exhaust pipe gasket
15. Exhaust muffler hanger

Catalytic converter removal steps

3. Harness cover
4. Heated oxygen sensor (rear) connector

<<A>> >>B<<

5. Heated oxygen sensor (rear)
6. Catalytic converter
14. Exhaust pipe gasket
16. Exhaust pipe gasket
17. Exhaust muffler hanger
18. Exhaust muffler hanger protector

Front exhaust pipe removal steps

- Cowl side trim (Refer to GROUP 52A, Trims [P.52A-11](#)).
- Footrest
- Floor console side cover (Refer to GROUP 52A, Front Floor Console Assembly [P.52A-9](#)).
- Turn up the passenger side floor carpet.
- Front floor backbone brace (Refer to GROUP 42, Loose Panel [P.42A-192](#)).
- 7. Heated oxygen sensor (front) connector
- 8. Heated oxygen sensor (front)
- 9. Front exhaust pipe
- 16. Exhaust pipe gasket
- 19. Seal ring

Front floor panel front heat protector and front floor panel rear heat protector removal steps

2. Center exhaust pipe
3. Harness cover

Front floor panel front heat protector and front floor panel rear heat protector removal steps (Continued)

4. Heated oxygen sensor (rear) connector
6. Catalytic converter
 - Cowl side trim (Refer to GROUP 52A, Trims [P.52A-11](#)).
 - Footrest
 - Floor console side cover (Refer to GROUP 52A, Front Floor Console Assembly [P.52A-9](#)).
 - Turn up the passenger side floor carpet.
 - Front floor backbone brace (Refer to GROUP 42, Loose Panel [P.42A-192](#)).
7. Heated oxygen sensor (front) connector
9. Front exhaust pipe
10. Seal ring
14. Exhaust pipe gasket
16. Exhaust pipe gasket
19. Seal ring
 - Propeller shaft assembly (Refer to GROUP 25, Propeller Shaft [P.25-6](#)).
- <> >>A<< 20. Rivet
- <> >>A<< 21. Front floor rear panel heat protector
22. Dash panel heat protector
- <> >>A<< 23. Rivet
24. Heated oxygen sensor clip
- <> >>A<< 25. Front floor front panel heat protector

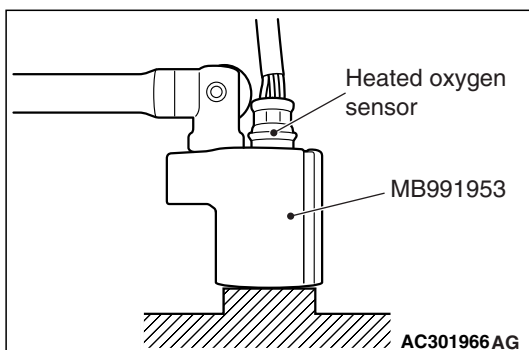
Required Special Tool:

- MB991953: Oxygen Sensor Wrench

REMOVAL SERVICE POINTS

<<A>> HEATED OXYGEN SENSOR REMOVAL

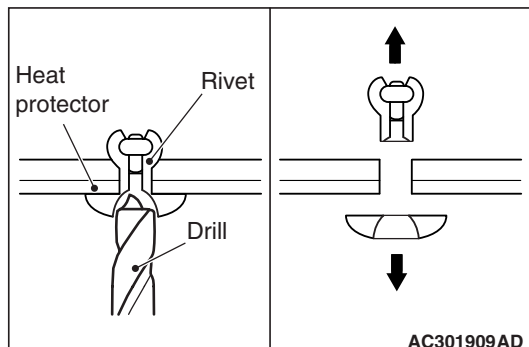
Use special tool MB991953 to remove the heated oxygen sensor.



<> RIVET/FRONT FLOOR REAR PANEL HEAT PROTECTOR/FRONT FLOOR FRONT PANEL HEAT PROTECTOR REMOVAL**⚠ CAUTION**

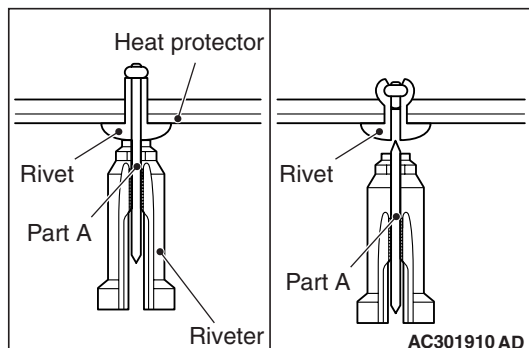
Be careful not to score the heat protector by drill.

1. Use a 6.0 / diameter / mm drill to make a hole in the flange of rivet as shown.
2. Break the rivet, and remove the rivet and heat protector.

**INSTALLATION SERVICE POINTS****>>A<< RIVET/FRONT FLOOR FRONT PANEL HEAT PROTECTOR/FRONT FLOOR REAR PANEL HEAT PROTECTOR/RIVET INSTALLATION**

Use a riveter to tighten the rivet by the following procedure.

1. Insert part A of rivet into the riveter, and insert the rivet into the assembling area.
2. Operate the tool's handle while pushing the flange surface of rivet by riveter. Part A of rivet is cut and the rivet is tightened.

**>>B<< HEATED OXYGEN SENSOR INSTALLATION**

Tighten the heated oxygen sensor to the specified torque by using special tool MB991953.

Tightening torque: 44 ± 5 N·m (33 ± 3 ft-lb)

